



ELSEVIER

Contents lists available at ScienceDirect

Journal of Ethnopharmacology

journal homepage: www.elsevier.com/locate/jethpharm



Arnebie Radix prevents atrial fibrillation in rats by ameliorating atrial remodeling and cardiac function

Qian Zhou^{a,1}, Bin Chen^{b,1}, Xiaodong Chen^{a,1}, Yue Wang^a, Jiawen Ji^a, Murat Kizaibek^d, Xindong Wang^a, Lixing Wu^a, Zhengli Hu^a, Xin Gao^a, Na Wu^a, Dan Huang^a, Xiaojin Xu^a, Wuguang Lu^a, Xueting Cai^a, Yang Yang^a, Juan Ye^a, Qingyun Wei^a, Jianping Shen^{a,*}, Peng Cao^{a,c,**}

^a Affiliated Hospital of Integrated Traditional Chinese and Western Medicine, Nanjing University of Chinese Medicine, Nanjing, 210028, Jiangsu, China

^b Nanjing Research Institute for Comprehensive Utilization of Wild Plants, Nanjing, 210042, Jiangsu, China

^c College of Pharmacy, Nanjing University of Chinese Medicine, Nanjing, 210023, Jiangsu, China

^d Traditional Kazakh Medicine Research Institute of Ili Kazakh Autonomous Prefecture, Traditional Chinese Medicine Hospital of Ili Kazakh Autonomous Prefecture, Yining, Xinjiang, 835000, China

ARTICLE INFO

Keywords:

Arnebie radix
Traditional Chinese medicine
Atrial fibrillation
Photochemistry
Cardiovascular disease

ABSTRACT

Ethnopharmacological relevance: *Arnebie Radix*, a common herbal medicine in China, is often utilized to treat blood-heat syndrome and has been reported to exert an effect on the heart.

Aim of the study: The combination of acetylcholine (Ach) and CaCl₂ has been widely used to induce atrial fibrillation (AF) in animals. However, whether *Arnebie Radix* displays any preventive action on Ach-CaCl₂ induced AF in rats remains uncertain. In our study, we attempted to investigate the protective effects of *Arnebie Radix* on Ach-CaCl₂ induced AF compared to amiodarone, which was employed as the positive control.

Materials and methods: To establish the AF model, SD rats were treated with a mixture of 0.1 mL/100 g Ach-CaCl₂ (60 µg/mL Ach and 10 mg/mL CaCl₂) by tail vein injection for 7 days. Rats were also given a gavage of *Arnebie Radix* (0.18 g/mL) one week before or concurrently with the establishment of the AF model. At the end of the experimental period, the induction, duration and timing of AF were monitored using electrocardiogram recordings. Left atrial tissues were stained to observe the level of fibrosis. Electrophysiological measurements were used to examine atrial size and function.

Results: In Ach-CaCl₂-induced AF rats, *Arnebie Radix* decreased AF induction, duration and susceptibility to AF. In addition, *Arnebie Radix* significantly reduced atrial fibrosis and inhibited atrial enlargement induced by Ach-CaCl₂. Moreover, there was an apparent improvement in cardiac function in the *Arnebie Radix*-treated group.

Conclusions: Our findings indicate that *Arnebie Radix* treatment can attenuate Ach-CaCl₂-induced atrial injury and serve as an effective therapeutic strategy for the treatment of AF in the future.

1. Introduction

Atrial fibrillation (AF) is similar to palpitation in traditional Chinese medicine (TCM). It is the most common cardiac arrhythmia and is characterized by disorganized, rapid, and irregular atrial activation with loss of atrial contraction and an irregular ventricular rate (Hung et al., 2016). The most serious disorder of atrial electrical activity is associated with regular displacement of the P-wave by the atrial fibrillary wave. In China, approximately 1.4% of men and 0.7% of

women have AF (Chen et al., 2016). The incidence of AF increases with age, and the estimated lifetime risk of developing AF is more than 5% in individuals older than 70 years of age. Antiarrhythmic drugs (sotalol, propafenone, flecainide, ibutilide and amiodarone) are commonly used to manage AF. Although effective, they are associated with limitations such as limited efficacy, adverse events, toxicity, and proarrhythmic potential (Angdisen et al., 2005; Camm, 2012).

The key pathogenic factors of atrial fibrillation are “stagnation” and “fire”. Treatment emphasizes the clearing away of heat and cooling the

* Corresponding author. Affiliated Hospital of Integrated Traditional Chinese and Western Medicine, Nanjing University of Chinese Medicine, 100#, Shizi Street, Hongshan Road, 210028, Nanjing, Jiangsu, China.

** Corresponding author. Affiliated Hospital of Integrated Traditional Chinese and Western Medicine, Nanjing University of Chinese Medicine, 138 Xianlin Avenue, 210023, Nanjing, Jiangsu, China.

E-mail addresses: shenjp@jsatcm.com (J. Shen), cao_peng@njucm.edu.cn (P. Cao).

¹ These authors contributed equally to this study.

blood (Wang, 2007). *Arnebieae Radix* (purple gromwell, the dried root of *Lithospermum erythrorhizon* Sieb. et Zucc, *Arnebia euchroma* (Royle) Johnston, *Arnebia guttata* Bunge and others) is a commonly used herbal medicine in China that is considered to be “cold” in nature. *Arnebieae Radix* acts on the heart and can eliminate blood-heat, suggesting its therapeutic effects on palpitation (Yang, 1998). Recently, the effect of *Arnebieae Radix* on cardiovascular system disorders have been studied (Ma et al., 2014). Shikonin is known to display diverse biological activities as an active ingredient of *Arnebieae Radix*. Shikonin has been proposed to have a potential role in reversing heart failure by reducing inflammation and apoptosis induced by endoplasmic reticulum stress (Yang et al., 2017a). It is also thought to have an effect on the inhibition of influenza and encephalitis B viruses from coronary heart disease (Xu and Xu, 2009). Because it contains the Lanthanum element, *Arnebieae Radix* might also be beneficial for lowering blood lipid levels (Xu and Xu, 2009). Such promising results suggest that *Arnebieae Radix* can be a candidate for developing new cardiovascular drugs. However, whether and how *Arnebieae Radix* affects AF remains uncertain. Considering the limited scientific information between *Arnebieae Radix* and AF, the purpose of the present study was to analyze the beneficial application of *Arnebieae Radix* in AF treatment *in vivo*, as well as the underlying mechanism.

2. Materials and methods

2.1. Chemicals and reagents

Standard shikonin compound (purity > 98%) was purchased from Cdmust Biotech Co., Ltd (Chengdu, China, batch number: 18040409); Amiodarone hydrochloride, acetylcholine chloride (Ach), calcium chloride (CaCl_2), analytical grade formic acid and ammonium formate were purchased from Sigma Chemical Co (St Louis MO). Amiodarone stock solution was prepared using ddH₂O at 50 mg/mL then heated to complete dissolution and stored at 4 °C. HPLC grade methanol was purchased from Merck (Darmstadt, Germany). Deionized water used throughout the experiments was purified with a Milli-Q academic water purification system (Millipore, Bedford, MA, USA). The other chemical reagents of analytical grade were purchased from local chemical suppliers.

2.2. Plant material

Arnebieae Radix [roots of the medicinal *Arnebia euchroma* (Royle) Johnston.(A.e.)] used for the preparation of *Arnebieae Radix* decoction was provided by the Jiangsu Province Hospital on Integration of Chinese and Western Medicine in February 2019 (Nanjing, China, batch number:180819) and identified by Professor Murat Kizaibek, Traditional Kazakh Medicine Research Institute.

2.3. Sample preparation

To prepare *Arnebieae Radix* decoction, plants were soaked in 10 vol of distilled water for 30 min, heated to boiling temperature and maintained at 100 °C for 30 min. The decoction was then filtered and the residual herbs were extracted with 8 vol of water in a second extraction. The filtrates obtained from the two extractions were condensed to 0.18 g/mL (i.e., each 0.18 g of raw material was finally concentrated to 1 mL of extract). The *Arnebieae Radix* decoction was packed separately and stored at −80 °C until use at room temperature.

For RP-HPLC method, the *Arnebieae Radix* decoction sample was treated with ultrasonic instrument for 30 min after adding 2-fold methanol into the sample. The temperature of the ultrasonic bath was kept at 25 ± 1 °C with running water. Subsequently, the extracts were filtered through a 0.45 μM PIFE syringe filter and subjected to RP-HPLC analysis. A standard shikonin solution of 1 mg/mL was prepared by dissolving accurately weighed quantity of shikonin in methanol and

stored at 4 °C.

For LC-MS analysis, RP-HPLC eluate within the retention time at 7.040 min was collected and compared to shikonin standard by MS spectra; 0.01 mg/mL of shikonin methanol solution was used as the standard solution.

2.4. RP-HPLC analysis

RP-HPLC analysis was carried out using a LaChrom Elite liquid chromatograph (Merck Hitachi, Japan) equipped with an autosampler and a column oven. Chromatographic separation was performed on a Gemini C₁₈ column (250 × 4.6 mm, 5 μm); the mobile phase was composed of methanol and 0.1% phosphate buffer (85:15, v:v). Flow rate and column temperature were maintained at 1.0 mL/min and 25 °C, respectively. The mobile phase was prepared fresh on the day of use, filtered through 0.45 μM nylon filter, and deaerated by sonication for 30 min. Detection was performed at 515 nm and injection volume was 10 μL.

2.5. Validation of the RP-HPLC method

2.5.1. Linearity

Calibration curve was generated using different concentrations of standard shikonin in the range of 0.4–1.6 mg/mL for the RP-HPLC method. Linearity was evaluated by linear regression analysis.

2.5.2. Precision

Precision was determined by repeatability (intra-day) and intermediate precision (inter-day). Intermediate precision was estimated by comparing the assays on three different days. Repeatability was assessed by analyzing six standard shikonin solutions of 1.0 mg/mL under the same operating conditions over a short time interval.

2.5.3. Accuracy

Accuracy was determined by recovery of known amounts of shikonin standard in triplicate. The criteria for data acceptability included accuracy within 85–115% (Wang et al., 2012).

2.5.4. Stability

Stability was assessed by analyzing the same sample solution at different times within 24 h at room temperature. The RSD values of the peak areas were all lower than 2.0%, indicating that it was safe to analyze the samples within 24 h.

2.5.5. Recovery

Recovery was studied in triplicate in standard shikonin solution at different concentration spiked with analytes. Recoveries were determined by comparing the peak area from the extracted sample to those for the reconstituted standard shikonin solution spiked with analytes.

2.6. LC-MS analysis

LC-MS analysis was performed on a Waters ACQUITY UPLC™ system (Waters Co., Milford, MA, USA). Chromatographic separation was carried out at 35 °C on Tris-Sal Plus C18 (250 mm × 4.6 mm, 5 μm). The flow rate was set at 1.0 mL/min with a split ratio of 3:2. The mobile phase was composed of 0.1% formic acid-10 mM/L ammonium formate (A) and 100% acetonitrile (B). Injection volume was 10 μL. Detections were performed using a tandem quadrupole mass spectrometer (Acquity TDQ™, Waters) in negative mode. Multiple-reaction-monitoring (MRM) scan mode was adopted for mass spectra. Ranges of m/z 287.1–218.1 for shikonin at collision energies of 30 eV were monitored. Spray voltage and capillary temperature were set at 3500 V and 350 °C. Analyte content was analyzed by the Masslynx V4.1 software (Waters Co., Milford, USA).

2.7. Animals care

All experimental procedures were approved by the Institutional Animal Care and Use Committee of the Jiangsu Province Institute of Traditional Chinese Medicine and were written following the ARRIVE guidelines. Experiments were performed in strict accordance with the National Institutes of Health (NIH) Guidelines for the Care and Use of Laboratory Animals. Male SD rats (8 weeks old, 250–300 g) were purchased from the Laboratory of Animal Science at the Chinese Academy of Medical Sciences (Beijing, China) and housed in a specific pathogen-free facility maintained at 20–22 °C in a 12-h light-dark cycle and provided with food and water *ad libitum*. Before the experiments, all rats were allowed to adapt to the conditions of the environment for 7 days.

2.8. Experimental design

Forty rats were assigned to four groups (n = 10 per group): 1) control group where each rat received 0.9% sodium chloride by both tail vein injection (i.v.) and oral administration (o.p.) for 7 days; 2) model group where each rat received a daily injection of a mixture containing 0.1 mL/100 g Ach-CaCl₂ (i.v.) for 7 days (Yang et al., 2017b; Zou et al., 2016); 3) *Arnebiae Radix* group where each rat received a daily injection containing a mixture of 0.1 mL/100 g Ach-CaCl₂ (i.v.) and *Arnebiae Radix* decoction (0.09 g/100 g) (o.p.) for 7 days; and 4) amiodarone group (n = 10) where each rat received a daily injection containing a mixture of 0.1 mL/100 g Ach-CaCl₂ (i.v.) and amiodarone (50 mg/kg) (o.p.) for 7 days. Amiodarone doses were based on published studies (Francisco et al., 2016; Shayeganpour et al., 2005). The dosage of *Arnebiae Radix* decoction was based on our preliminary studies and human equivalent doses of raw herb used according to the Pharmacopoeia of China with 10 g (Commission, 2015).

2.9. ECG collection

On the 8th day, rats from each group were anesthetized using 10% chloral hydrate. Saline was then given to control rats while others were given a mixture of Ach-CaCl₂ solution via the tail vein. Lead II surface electrocardiogram (ECG) monitoring was performed with 25-gauge subcutaneous needle electrodes. ECG signals were filtered (0.5–150 Hz) and amplified (20 mm/mV) with an FX8322 electrocardiographic system (FUKUDA DENSHI, Japan). The sweep of ECG paper was performed at 50 mm/s. Analog ECG recordings at baseline were made at 25–100 mm/s paper speed and 1 mm/mV amplitude. P-wave duration was acquired and analyzed from the time when administration began.

2.10. Electrophysiological measurements

After ECG detection, six customized electrodes were sutured in the right atrium (RA), left atrium (LA), and four pulmonary veins, respectively. After induction of anesthesia, rats were placed in a decubitus supine position on a heat pad. After pre-warming, echo transmission gel was applied to the hairless chest of rats, and transthoracic echocardiography was performed using GE ViVid-q ultrasound systems (GE Systems, Hayozma, Tirat Carmel, Israel). The two-dimensional guided M-mode image was obtained in the short-axis view at the level of the papillary muscles using a 13-MHz linear transducer (8L-RS). Two-dimensional imaging was acquired at a minimum depth setting of 2 cm. M-mode spectra were set at a sweep speed of 200 mm/s to determine the following parameters: LV end-systolic diameter (LVESD) and LV end-diastolic diameter (LVEDD). To reflect cardiac function, fractional shortening (FS) was calculated using the formula: $FS\% = [(LVEDD - LVESD) / LVEDD] \times 100\%$. Left atrial diameter (LAD) and LA area were measured at the level of the aortic sinus in the parasternal long-axis view using the Mode tracing method. All observers were blinded to the rats that they would assess. Every parameter was measured three times

and the average values were calculated.

2.11. Histological examination

The heart tissue samples were isolated from rats at the end of our study and fixed in 4% paraformaldehyde phosphate buffered saline (PBS) solution overnight. The tissues were sectioned and dehydrated through a series of graded ethanol, cleared in xylene, and embedded in paraffin block. The blocks were then cut into 4 μm-thick sections and mounted on a glass slide. The deparaffinized tissue sections were stained with Masson's trichrome blue and Sirius red staining to visualize atrial fibrosis. Immunohistochemistry was performed for α-SMA and Collagen I. A positive reaction primarily referred to the brown-yellow granules in the atrial tissues, and the number and darkness of granules bespoke the expression levels of target proteins. The images of stained sections were captured with a Nikon Eclipse Ti-S microscope with a digital camera. The calculations and analysis of the stained images were performed using Image-Pro Plus 6.0 software (Media Cybernetics, Rockville, MD, USA).

2.12. Statistical analysis

Results are expressed as mean ± standard error of the mean (mean ± SEM). Student's-test for non-paired replicates was used to identify statistically significant differences between treatment means. A difference was considered significant when the P value was < 0.05.

3. Results

3.1. Phytochemical analysis of *Arnebiae Radix* decoction by RP-HPLC

RP-HPLC analysis was conducted to evaluate the content of shikonin for *Arnebiae Radix* decoction samples. An external standard method was applied and validated in terms of linearity, precision, accuracy, stability, and recovery. A typical base peak intensity chromatogram of shikonin was observed at 7.042 min in standard shikonin (Fig. 1-A) and 7.040 min in *Arnebiae Radix* decoction sample (Fig. 1-B). Based on the equation of the calibration curve: $y = 3051.4x + 1239.9$ ($r^2 = 0.9997$, where x = standard shikonin concentration and y = peak area), the content of total shikonin in *Arnebiae Radix* decoction sample was 1.335%. Such result was promising and fully complied with the requirement of the Chinese pharmacopoeia (Commission, 2015). The eluate of *Arnebiae Radix* decoction samples within the retention time at 7.040 min was confirmed by comparison to standard shikonin using LC-MS analysis (Supplementary Fig. S1).

3.2. *Arnebiae Radix* decoction reduces the induction, susceptibility, and duration of AF

The absence of P-waves and appearance of f-waves on the ECG indicated the successful induction of AF (Fig. 2-A). This induction was approximately 88.89% in model group rats. In contrast, a decrease was observed in the *Arnebiae Radix* (75%) and amiodarone (70%) treated groups (Fig. 2-B). Duration is the length of time that AF lasts while induction time refers to the time when AF is triggered (Zhao et al., 2017). On the 8th day, duration for the model group was approximately 8.18 ± 0.31 s, this was significantly longer than the amiodarone-treated group (2.12 ± 0.24 s) and *Arnebiae Radix* group (6.96 ± 1.89 s) (Fig. 2-C). In addition, an evident prolonging of the induction time of AF was observed in the *Arnebiae Radix* group (7.03 ± 0.75 s) compared to the model group (5.84 ± 0.46 s), as it is relatively harder to induce AF in *Arnebiae Radix* group (Fig. 2-D). These results suggested that *Arnebiae Radix* reduced the occurrence and downplayed the susceptibility of AF.

(A) Representative images of ECG recordings of normal sinus rhythm and atrial fibrillation episodes. The inducibility of AF in the

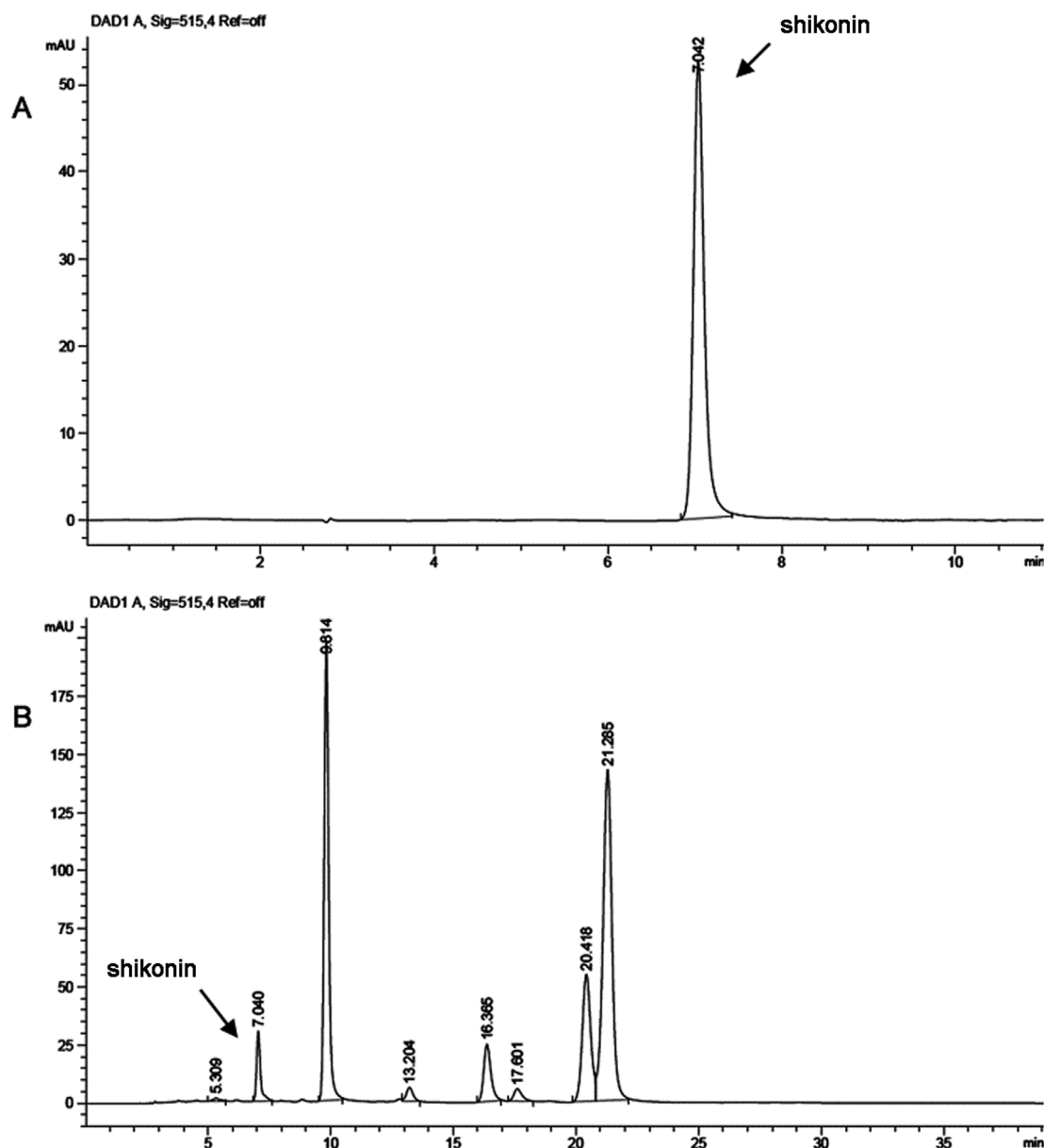


Fig. 1. RP-HPLC chromatogram analysis was set at 515 nm of shikonin and an *Arnebie Radix* decoction sample.

four groups is detailed in (B). Mean atrial fibrillation duration in the four groups is detailed in (C). The time that triggered atrial fibrillation in the four groups. Each value is presented as mean $\pm$ SEM, $n = 10$. * $P < 0.05$, ** $P < 0.01$ vs. control group; # $P < 0.05$, ## $P < 0.01$ vs. Ach- CaCl_2 treatment model group.

3.3. *Arnebie Radix* decoction attenuates atrial fibrosis triggered by AF

In the clinical setting, AF is associated with interstitial fibrosis, which correlates with structural remodeling and plays a role in sustaining arrhythmia (Klesen et al., 2018; Zhang et al., 2018). Masson staining was therefore adopted to evaluate the extent of interstitial fibrosis in atrial tissues (fibers in the tissues appeared blue and atrial myocytes appeared red). The extent of interstitial fibrosis area was defined as: (blue area/[red area + blue area]) (Yang, 1998; Zhang et al., 2018). As shown in Fig. 3-A, the level of fibrosis increased substantially and the atrial tissues were significantly separated by fibers in the model group. Compared to the model group, an evident degree of alleviation was observed in the *Arnebie Radix* and amiodarone groups.

Sirius red staining was used to evaluate the level of type I collagen and type III collagen in atrial tissues (Ma et al., 2017). Type I and III collagens were significantly increased in the atrial area in the model group (Fig. 3-B) and a certain degree of deposition was observed in the *Arnebie Radix* and amiodarone groups as shown by the decrease in the positively-stained area (Fig. 3-B). We detected the levels of α -SMA and collagen type I, two key markers of fibrosis, by immunohistochemical staining. Results showed that Ach- CaCl_2 significantly elevated the expression of α -SMA and collagen type I compared to those of the control (Fig. 3-C and D). Apparently, *Arnebie Radix* and amiodarone significantly down regulated the expression of these fibrosis markers (Fig. 3-C and D). Collectively, *Arnebie Radix* suppressed atrial fibrosis in AF rats.

(A) Left, atrial sections from rats were stained with Masson's trichrome, and the blue color referred to the collagen accumulation levels. Right, the interstitial fibrosis area was quantified. (B) Left, atrial sections from rats were stained through Sirius Red staining were exhibited. Right, the fibrosis levels were calculated. Immunohistochemical analysis was carried out to calculate α -SMA (C) and Collagen I levels.

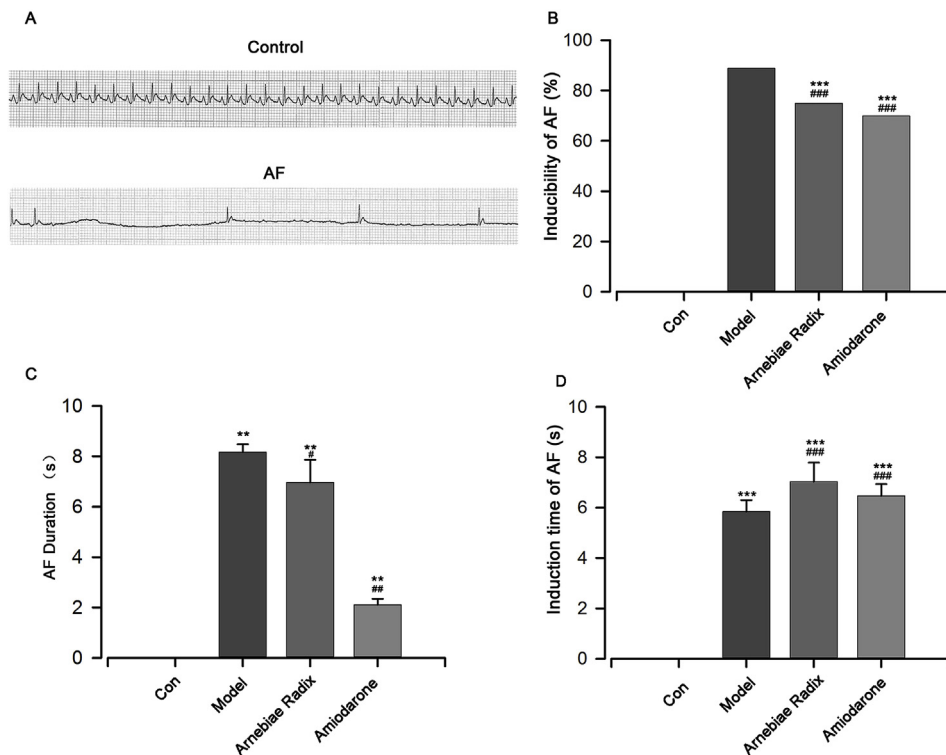


Fig. 2. *Arnebiae Radix* decoction reduces the induction, susceptibility and duration of AF.

Representative images (Left) and the quantified results (Right) are displayed. Each value is presented as mean $\pm$ SEM, $n = 10$. * $P < 0.05$, ** $P < 0.01$ vs. control group; # $P < 0.05$, ## $P < 0.01$ vs. Ach- CaCl_2 treatment model group.

3.4. *Arnebiae Radix* decoction inhibits LAD and LA area, and improves cardiac function

As cardiac function and LAD are related to AF (Qiu et al., 2018), we used echocardiography to measure these indices at the end of the treatment period (Fig. 4-A). As shown in Fig. 4, the FS values in the model group were reduced, and LAD and LA areas were augmented compared to those in the control group. *Arnebiae Radix* treatment

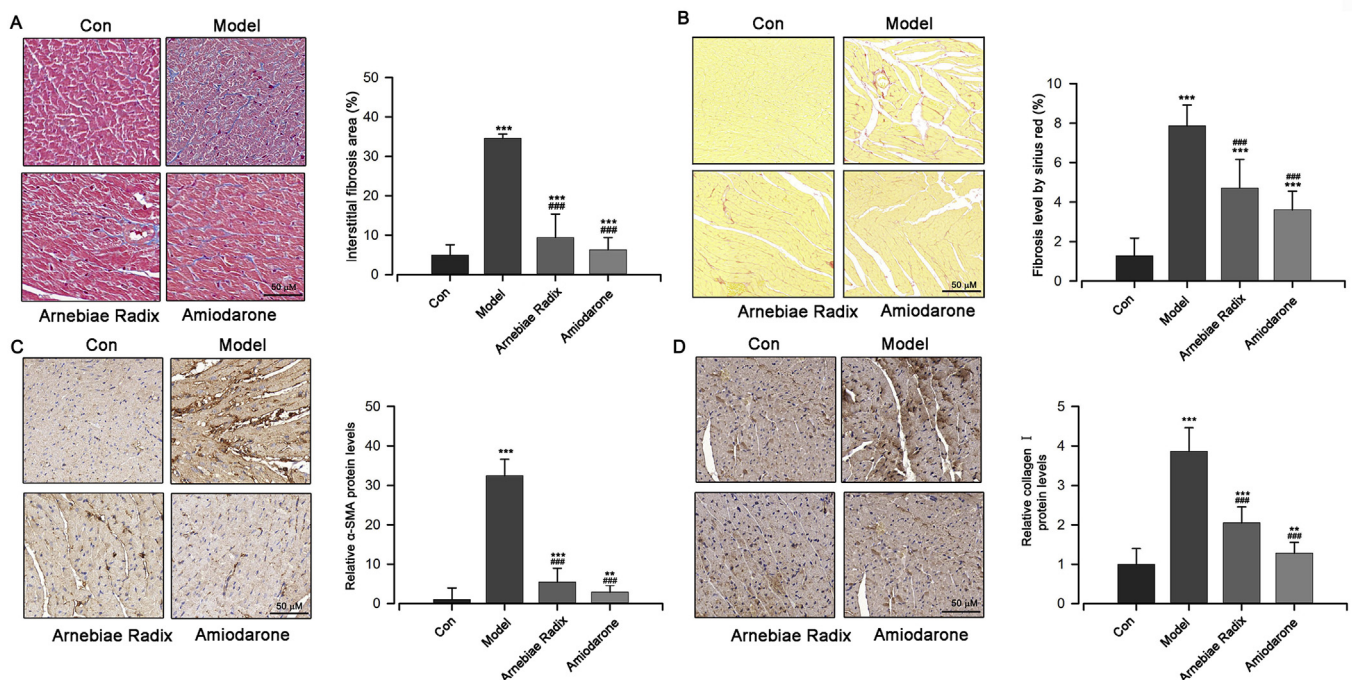


Fig. 3. *Arnebiae Radix* decoction attenuates atrial fibrosis triggered by AF.

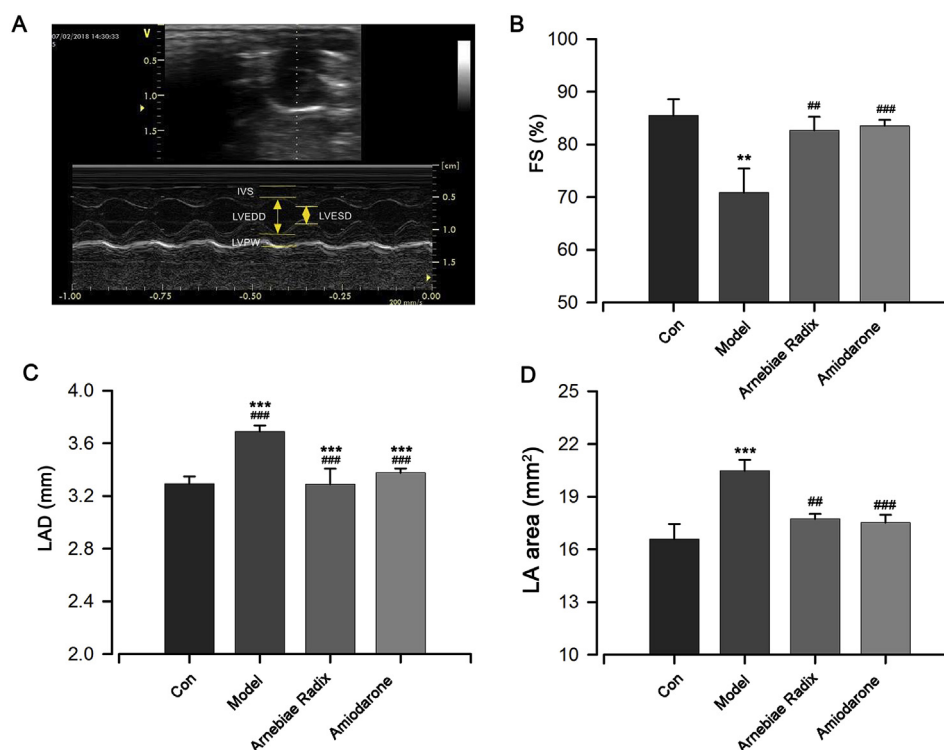


Fig. 4. *Arnebie Radix* decoction inhibits LAD and LA area, and improves cardiac function.

reduced the deterioration in cardiac function (Fig. 4-B) and atrial enlargement (LAD and LA area values). Therefore, we revealed that *Arnebie Radix* displays a protective effect on cardiac function and atrial size. Amiodarone significantly restrained atrial size expansion (Fig. 4-C and D) and improved cardiac function (Fig. 4-B).

(A) Representative echocardiography images in parasternal short axis view. (B) FS, (C) LAD, (D) LA area. IVS: Inter ventricular septum thickness in diastole; LVEDD: Left ventricular end-diastolic dimension; LVESD: left ventricular end-systolic dimension; LVPW: Left ventricular thickness wall thickness degree; FS: fractional shortening; LAD: Left atrial diameter; LA area: Left atrial diameter area; Each value is presented as mean $\pm$ SEM, $n = 10$. * $P < 0.05$, ** $P < 0.01$ vs. control group; # $P < 0.05$, ## $P < 0.01$ vs. Ach-CaCl₂ treatment model group.

4. Discussion

The main mechanisms contributing to the initiation and maintenance of AF include atrial electrical and structural remodeling (Yang et al., 2016). Although the Ach-CaCl₂ induced AF model is not widely used as the model triggered by atrial burst pacing (Yang et al., 2016; Yoon et al., 2013b), it has been successfully established in rats (Guo et al., 2010; Yang et al., 2017b; Zou et al., 2016). Recent observations indicated that paroxysmal AF is initiated from the pulmonary area. Ach could trigger re-entrance circuit(s) that is/are stabilized around the pulmonary veins where a high concentration of Ach could result in AF development (Wyse and Gersh Bernard, 2004). Besides, it is widely recognized that Ach activated-potassium channels (I_{KAch}) can be pre-disposed to AF and lack of I_{KAch} can prevent AF (Kovoor et al., 2001). A rapid increase in CaCl₂ concentration could lead to severe hyperkalemia causing calcium influx. Therefore, combining CaCl₂ and Ach could result in partial blockage of conduction and inconsistent impulse diffusion, inducing AF and reflecting the natural pathological process of AF (Chen et al., 2009; Guo et al., 2010; Zou et al., 2016). Consistent with the previous study, our result demonstrated the typical AF wave in the ECG, atrial structural remodeling (atrial enlargement and fibrosis) and decreased cardiac function in rats with AF (Figs. 2-4).

AF is initially induced by an abnormal electrical activity and is maintained in the atrium with irregular re-entry (Yoon et al., 2013a). Researchers have demonstrated that maintenance of rhythm plays a key role in AF (Deng et al., 2010). Amiodarone is one of the most commonly used antiarrhythmic agents for the conversion of AF; however, is limited because of its effectiveness and delayed onset of action. In one study, amiodarone displayed conversion in only 5% of patients within the first 90 min in recent-onset AF (Camm et al., 2011) and approximately 20%–25% in others (Martínez-Marcos et al., 2000). Secondly, median time for the reversion of acute AF to sinus rhythm by amiodarone required several hours (Khan et al., 2003; Martínez-Marcos et al., 2000). Thirdly, due to its proarrhythmic potential and a variety of adverse effects, few patients tolerated amiodarone based on long-term concern (Smith et al., 1986). *Arnebie Radix*, a commonly used herbal medicine in China with a rather cool character, has long been used to cool blood and for detoxification and anti-inflammation (Kim et al., 2019). Clinically, oral administration of a multi-herb decoction containing *Arnebie Radix* with a dosage of 15 g per day for 2 months reported no adverse events (May et al., 2012). Meanwhile, repeated oral administration that is 16 times higher than the effective dose of *Arnebie Radix* aqueous extract for 12 weeks resulted in no remarkable toxic signs or parameter variations in experimental studies in humans (Kim et al., 2019). Therefore, a much greater long-term tolerance and safety of *Arnebie Radix* could be deduced.

In this study, AF was triggered by injection with the mixture of Ach-CaCl₂; injection with saline did not affect ECG wave in control rats (Fig. 2-A). AF inducibility and duration were significantly increased in model rats (Fig. 2-B and C) which was closely related to the clinical condition associated with AF (Chen et al., 2009). *Arnebie Radix* has an anti-AF effect by shortening inducibility and duration similar to that of amiodarone (Fig. 2-B and C). Interestingly, *Arnebie Radix* treatment could better prolong the induction time of AF, thereby lowering AF susceptibility (Fig. 2-D). AF susceptibility depends on the electrical and structural remodeling of atrium (XU et al., 2012). Therefore, we infer that *Arnebie Radix* might have an anti-AF effect based on its amelioration of atrial remodeling.

Atrial fibrosis is the most prominent feature of atrial structural remodeling that is associated with AF (Wang et al., 2015). Atrial fibrosis contributes to an important pathological basis for the occurrence and maintenance of AF. Inhibiting the process of atrial fibrosis may be an important target for AF intervention (Zhang et al., 2018). In this study, Masson's trichrome stain indicated interstitial fibrosis in Ach-CaCl₂ treated atrial tissues, and both *Arnebiae Radix* and Amiodarone treatment significantly reduced atrial fibrosis (Fig. 3-A). Consistently, sirius red staining revealed an increased level of type I and collagen III accumulation, which could be reversed by *Arnebiae Radix* and amiodarone, respectively (Fig. 3-B). Additionally, fibrosis-related markers of α -SMA and Collagen type I, which have been reported to play a key role in AF formation, were reversed by co-treatment with *Arnebiae Radix* or amiodarone as observed by immunohistochemical analysis (Fig. 3-A and B). The above findings support the anti-AF role of *Arnebiae Radix* in preventing atrial fibrosis.

LA enlargement has been identified as an independent risk factor for AF because patients are more prone to paroxysmal AF if LAD is increased (Cui et al., 2008). LA enlargement also correlates with atrial fibrosis, which is associated with the formation of AF and is difficult to reverse (Cui et al., 2008). In our study, we revealed that 7 days of injection with Ach-CaCl₂ pathologically resulted in atrial enlargement with marked increase in LAD, and LA area (Fig. 4-C). These parameter changes were linked to decreased FS in the model group (Fig. 4-B). Similar to amiodarone, treatment with *Arnebiae Radix* improved LA, LAD and FS (Fig. 4B–D) and similar to the above result for AF induction, atrial enlargement was better reversed in the *Arnebiae Radix* treated group than amiodarone treated group (Fig. 4-C and D). Taken together, these results demonstrate that *Arnebiae Radix* display anti-AF efficiency by delaying atrial remodeling.

5. Conclusion

In conclusion, the present study confirmed the protective effect of *Arnebiae Radix* against AF and demonstrated that the anti-AF effects may be associated with (a) reduced induction, susceptibility, and duration; (b) ameliorated remodeling; and (c) improved cardiac function. These findings may serve as evidence for the clinical choice of *Arnebiae Radix* in AF, and may provide new ideas for the clinical treatment of AF. A summary of our findings is shown in Graphical Abstracts.

Author contributions statement

Q.Z. obtained funding, conceived, designed, and acquired the data, performed the analysis and revised of the data; B.C. designed, performed the phytochemical analysis and interpretation of the data; J.J. and Y.W. acquired the data, and performed the analysis; M.K. identified the medicinal plants; X.C. developed the methodology, and wrote the manuscript; X.W., L.W., Z.H., X.G., N.W., D.H., X.X., W.L., and X.C., supported several experiments; Y.Y., J.Y., and Q.W. performed the statistical analysis and revised the manuscript; P.C. provided technical support, proof-read the manuscript; J.S. obtained funding, conceived, designed, and revised the manuscript. All authors read and approved the final manuscript.

Declarations of competing interest

None.

Acknowledgements

This study was supported by grants from the National Natural Science Foundation of China (grant number 81603332, 81873274), Natural Science Foundation of Jiangsu Province, China (grant number BK20161082), Postgraduate Research & Practice Innovation Program of

Jiangsu Province, China (grant number KYCX191280), Nanjing Science and Technology Plan, China (grant number 201803065) and the Science and Technology Project of Jiangsu Traditional Chinese Medicine Bureau, China (grant number FY201806).

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jep.2019.112317>.

References

- Angdisen, J., Moore, V.D., Cline, J.M., Payne, R.M., Ibdah, J.A., 2005. Mitochondrial trifunctional protein defects: molecular basis and novel therapeutic approaches. *Curr. Drug Targets - Immune, Endocr. Metab. Disord.* 5 (1), 27–40.
- Camm, A.J., Capucci, A., Hohnloser, S.H., Torp-Pedersen, C., Van Gelder, I.C., Mangal, B., Beatch, G., 2011. A randomized active-controlled study comparing the efficacy and safety of vernakalant to amiodarone in recent-onset atrial fibrillation. *J. Am. Coll. Cardiol.* 57 (3), 313–321.
- Camm, J., 2012. Antiarrhythmic drugs for the maintenance of sinus rhythm: risks and benefits. *Int. J. Cardiol.* 155 (3), 362–371.
- Chen, C., Gong, T., Kou, Y., 2009. Establishment of atrial fibrillation animal model in rat. *Lab. Anim. Sci.* 26 (3), 1–4.
- Chen, G., Wei, B., Wang, J., Feng, B., Li, Z., Zhang, Z., He, Q., 2016. Shensongyangxin capsules for paroxysmal atrial fibrillation: a systematic review of randomized clinical trials. *PLoS One* 11 (3), e0151880.
- Commission, C.P., 2015. Pharmacopoeia of the People's Republic of China: the Fourth Division of 2015 Edition. Chemical and Technologic Press, Beijing.
- Cui, Q., Zhang, W., Wang, H., Sun, X., Yang, H., Meng, X., Zhang, Y., 2008. Left and right atrial size and the occurrence predictors in patients with paroxysmal atrial fibrillation. *Int. J. Cardiol.* 130 (1), 69–71.
- Deng, M., Sui, X.Q., Zhu, S.B., Ma, W., Xu, Y., Chen, Z.M., 2010. Clinical observation on the treatment of atrial fibrillation with amiodarone combined with Shenmai Injection (参麦注射液). *Chin. J. Integr. Med.* 16 (5), 453–456.
- Francisco, A., Dias, N.C., Menezes, M., Guimaraes, T., Silva, G., Placido, R., Magalhães, A., Inácio, C., Pinto, F.J., Coutinho, C.A., 2016. Prognostic significance of dysrhythmias in patients with familial amyloid polyneuropathy. *Eur. Heart J.* 37, 683–684.
- Guo, X., Chen, C.L., Yang, Q., Yin, Y.M., You, Q.D., Tang, Y.Q., 2010. Effects of a novel class III antiarrhythmic agent, CPUY11018, on rat atrial fibrillation. *Drug Dev. Res.* 71 (5), 303–312.
- Hung, Y.C., Cheng, Y.C., Muo, C.H., Chiu, H.E., Liu, C.T., Hu, W.L., 2016. Adjuvant Chinese herbal products for preventing ischemic stroke in patients with atrial fibrillation. *PLoS One* 11 (7), e0159333.
- Khan, I.A., Mehta, N.J., Gowda, R.M., 2003. Amiodarone for pharmacological cardioversion of recent-onset atrial fibrillation. *Int. J. Cardiol.* 89 (2), 239–248.
- Kim, N.S., Shin, S., Park, H.-S., Kwon, H.-J., Son, H.-Y., Bang, O.-S., 2019. Sub-chronic oral toxicity of the aqueous extract of lithospermum radix in Fischer 344 rats. *J. Ethnopharmacol.* 235, 406–414.
- Klesen, A., Jakob, D., Emig, R., Kohl, P., Ravens, U., Peyronnet, R., 2018. Cardiac fibroblasts: active players in (atrial) electrophysiology? *Herzschrittmachertherap. Elektrophysiol.* 29 (1), 62–69.
- Kovoor, P., Wickman, K., Maguire, C.T., Pu, W., Gehrmann, J., Berul, C.I., Clapham, D.E., 2001. Evaluation of the role of I_{KACH} in atrial fibrillation using a mouse knockout model. *J. Am. Coll. Cardiol.* 37 (8), 2136–2143.
- Ma, J., Li, Z.Y., Liang, X.P., Guo, C.X., Lu, P.P., Ma, L.H., 2017. Xinfuli Granule improves post-myocardial infarction ventricular remodeling and myocardial fibrosis in rats by regulating TGF- β /Smads signaling pathway. *J. Geriatr. Cardiol.* 14 (5), 301–307.
- Ma, Z.Q., Hu, H., He, T.T., Guo, H., Zhang, M.Y., Chen, M.W., Wang, Y.T., 2014. An assessment of traditional Uighur medicine in current Xinjiang region (China). *Afr. J. Tradit., Complement. Altern. Med.* 11 (2), 301–314.
- Martínez-Marcos, F.J., García-Garmendia, J.L., Ortega-Carpio, A., Fernández-Gómez, J.M., Santos, J.M.a., Camacho, C., 2000. Comparison of intravenous flecainide, propafenone, and amiodarone for conversion of acute atrial fibrillation to sinus rhythm. *Am. J. Cardiol.* 86 (9), 950–953.
- May, B.H., Zhang, A.L., Zhou, W., Lu, C.-j., Deng, S., Xue, C.C.L., 2012. Oral herbal medicines for psoriasis: a review of clinical studies. *Chin. J. Integr. Med.* 18 (3), 172–178.
- Qiu, H., Liu, W., Lan, T., Pan, W., Chen, X., Wu, H., Xu, D., 2018. Salvianolate reduces atrial fibrillation through suppressing atrial interstitial fibrosis by inhibiting TGF- β /Smad 2/3 and TXNIP/NLRP3 inflammasome signaling pathways in post-MI rats. *Phytomedicine* 51, 255–265.
- Shayeganpour, A., Jun, A.S., Brocks, D.R., 2005. Pharmacokinetics of Amiodarone in hyperlipidemic and simulated high fat-meal rat models. *Biopharm Drug Dispos.* 26 (6), 249–257.
- Smith, W.M., Lubbe, W.F., Whitlock, R.M., Mercer, J., Rutherford, J.D., Roche, A.H., 1986. Long-term tolerance of amiodarone treatment for cardiac arrhythmias. *Am. J. Cardiol.* 57 (15), 1288–1293.
- Wang, Q., 2007. Yi Lin Gai Cuo: Correcting the Errors in the Forest of Medicine. Blue Poppy Enterprises, Inc.
- Wang, R., Yi, X., Li, X., Jiang, X., 2015. Fibroblast growth factor-21 is positively associated with atrial fibrosis in atrial fibrillation patients with rheumatic heart disease.

- Int. J. Clin. Exp. Pathol. 8 (11), 14901.
- Wang, Y., Ding, C., Wu, C., Du, K., Zhang, J., Qin, H., Hou, J., Du, G., 2012. HPLC-MS and HPLC-MS/MS analysis of seven active constituents of Xiao-Xu-Ming decoction and application to a pharmacokinetic study after oral administration to rat. *Acta Pharm. Sin. B* 2 (2), 188–197.
- Wyse, D.G., Gersh Bernard, J., 2004. Atrial fibrillation: a perspective. *Circulation* 109 (25), 3089–3095.
- Xu, D., Murakoshi, N., Igarashi, M., Hirayama, A., Ito, Y., Seo, Y., Tada, H., Aonuma, K., 2012. PPAR- γ activator pioglitazone prevents age-related atrial fibrillation susceptibility by improving antioxidant capacity and reducing apoptosis in a rat model. *J. Cardiovasc. Electrophysiol.* 23 (2), 209–217.
- Xu, H., Xu, H.-E., 2009. Analysis of trace elements in Chinese therapeutic foods and herbs. *Am. J. Chin. Med.* 37 (04), 625–638.
- Yang, J., Wang, Z., Chen, D.L., 2017a. Shikonin ameliorates isoproterenol (ISO)-induced myocardial damage through suppressing fibrosis, inflammation, apoptosis and ER stress. *Biomed. Pharmacother.* 93, 1343–1357.
- Yang, Q., Lv, Q., Feng, M., Liu, M., Feng, Y., Lin, S., Yang, J., Hu, J., 2017b. Taurine Prevents the Electrical Remodeling in Ach-CaCl₂ Induced Atrial Fibrillation in Rats. Springer Netherlands, Dordrecht, pp. 821–830.
- Yang, Q., Qi, X., Dang, Y., Li, Y., Song, X., Hao, X., 2016. Effects of atorvastatin on atrial remodeling in a rabbit model of atrial fibrillation produced by rapid atrial pacing. *BMC Cardiovasc. Disord.* 16 (1), 142.
- Yang, S., 1998. The Divine Farmer's Materia Medica: a Translation of the Shen Nong Ben Cao Jing. Blue Poppy Enterprises, Inc.
- Yoon, N., Cho, J.G., Kim, K.H., Park, K.H., Sim, D.S., Yoon, H.J., Hong, Y.J., Park, H.W., Kim, J.H., Ahn, Y., Jeong, M.H., Park, J.C., 2013a. Beneficial effects of an angiotensin-II receptor blocker on structural atrial reverse-remodeling in a rat model of ischemic heart failure. *Exp. Ther. Med.* 5 (4), 1009–1016.
- Yoon, N., Kim, K.H., Park, K.H., Sim, D.S., Yoon, H.J., Hong, Y.J., Park, H.W., Kim, J.H., Ahn, Y., Jeong, M.H., Cho, J.G., Park, J.C., 2013b. Preventive effects of the angiotensin-II receptor blocker on atrial remodeling in an ischemic heart failure model of rats. *Korean Circ. J.* 43 (10), 686–693.
- Zhang, K., Ma, Z., Wang, W., Liu, R., Zhang, Y., Yuan, M., Li, G., 2018. Effect of doxycycline on chronic intermittent hypoxia-induced atrial remodeling in rats. *Herz* 1–8.
- Zhao, H.Y., Zhang, S.D., Zhang, K., Wang, X., Zhao, Q.Y., Zhang, S.J., Dai, Z.X., Qian, Y.S., Zhang, Y.J., Wei, H.T., Tang, Y.H., Huang, C.X., 2017. Effect of shensong yangxin on the progression of paroxysmal atrial fibrillation is correlated with regulation of autonomic nerve activity. *Chin. Med. J. (Engl)* 130 (2), 171–178.
- Zou, D., Geng, N., Chen, Y., Ren, L., Liu, X., Wan, J., Guo, S., Wang, S., 2016. Ranolazine improves oxidative stress and mitochondrial function in the atrium of acetylcholine-CaCl₂ induced atrial fibrillation rats. *Life Sci.* 156, 7–14.

Update

Journal of Ethnopharmacology

Volume 340, Issue , 31 January 2025, Page

DOI: <https://doi.org/10.1016/j.jep.2024.119264>



Corrigendum to “*Arnebiae Radix* prevents atrial fibrillation in rats by ameliorating atrial remodeling and cardiac function” [J. Ethnopharmacol. 248 (2020) 112317]

Qian Zhou^{a,1}, Bin Chen^{b,1}, Xiaodong Chen^{a,1}, Yue Wang^a, Jiawen Ji^a, Murat Kizaibek^d, Xindong Wang^a, Lixing Wu^a, Zhengli Hu^a, Xin Gao^a, Na Wu^a, Dan Huang^a, Xiaojin Xu^a, Wuguang Lu^a, Xueting Cai^a, Yang Yang^a, Juan Ye^a, Qingyun Wei^a, Jianping Shen^{a,*}, Peng Cao^{a,c,**}

^a Affiliated Hospital of Integrated Traditional Chinese and Western Medicine, Nanjing University of Chinese Medicine, Nanjing, 210028, Jiangsu, China

^b Nanjing Research Institute for Comprehensive Utilization of Wild Plants, Nanjing, 210042, Jiangsu, China

^c College of Pharmacy, Nanjing University of Chinese Medicine, Nanjing, 210023, Jiangsu, China

^d Traditional Kazakh Medicine Research Institute of Ili Kazakh Autonomous Prefecture, Traditional Chinese Medicine Hospital of Ili Kazakh Autonomous Prefecture, Yining, Xinjiang, 835000, China

The authors regret there is an error in the collagen I staining of the Amiodarone group of Fig. 3D. It was found that the error occurred during the process of image pasting and creating a uniform layout for the different groups. To rectify this issue, the authors have rectified the image based on the original raw data and have provided the uncropped

raw image used in Fig. 3D. The corrected Fig. 3D and raw data is displayed below. The corrections made to the image in this corrigendum do not affect the overall findings and conclusions of this study.

DOI of original article: <https://doi.org/10.1016/j.jep.2019.112317>.

* Corresponding author.

** Corresponding author. College of Pharmacy, Nanjing University of Chinese Medicine, Nanjing, 210023, Jiangsu, China.

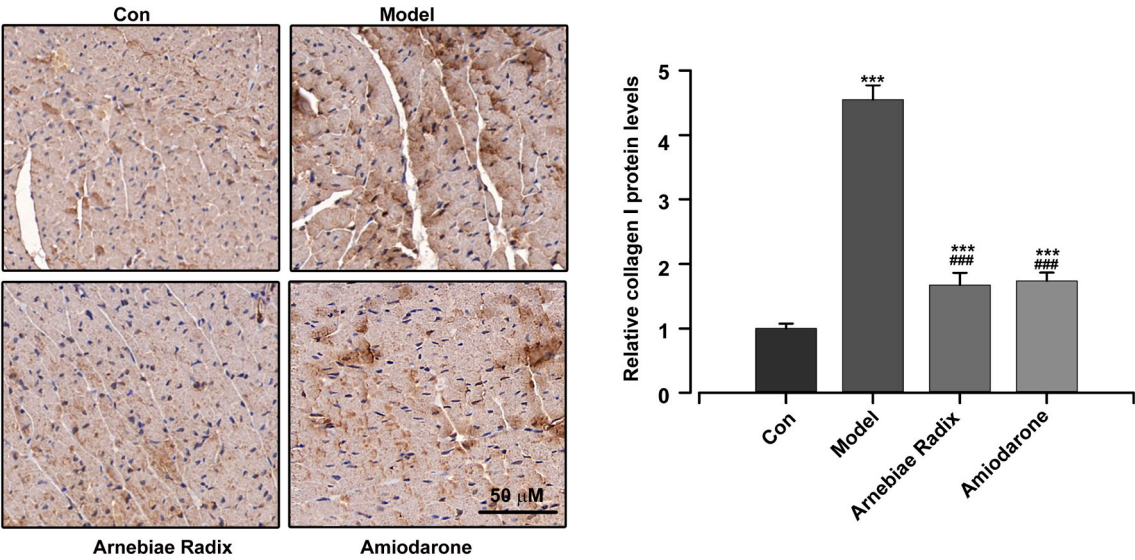
E-mail addresses: shenjp@jsatcm.com (J. Shen), cao_peng@njucm.edu.cn (P. Cao).

¹ These authors contributed equally to this study.

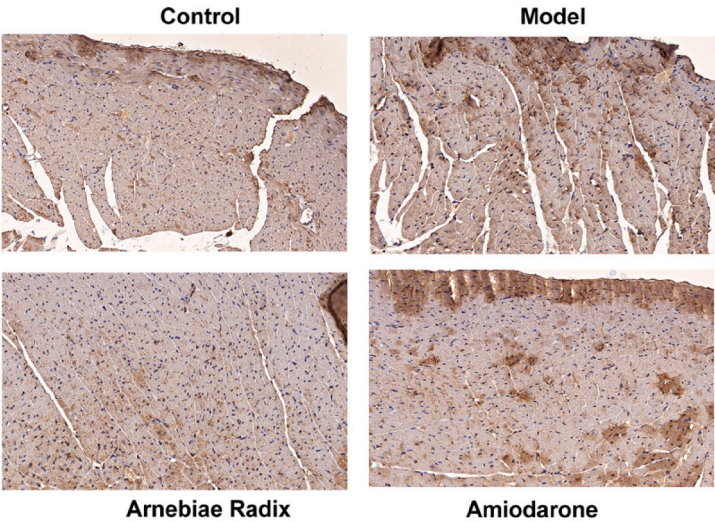
<https://doi.org/10.1016/j.jep.2024.119264>

Available online 22 December 2024

0378-8741/© 2024 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.



The correctd Figure 3D



The uncropped raw data used in Figure 3D

The authors would like to apologise for any inconvenience caused.